

Simple Adjustable Staircase Waveform Generator

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Staircase signals find many applications in the field of electronics, such as in television systems, telemetry, and analogue and digital communications. Here is a circuit illustrating the basic principles of staircase waveform generator. Advantages of this circuit are: simple, low-cost, no complex programming, easy-to-assemble on a breadboard and adjustable staircase waveform outputs of variable step sizes and durations.

Circuit and working

The circuit diagram of a simple adjustable staircase waveform generator is shown in Fig. 1.

IC1 (74HC14) is a hex inverter Schmitt trigger that generates oscillatory pulse (square wave) and gives an output clock at pin 14 of IC2 (74HC93),

which is a 4-bit binary ripple counter. Pins 8, 9 and 12 of IC2 are connected to select pins of IC3 (74HC4051), which is

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Semiconductors:	
IC1	- 74HC14 hex inverter Schmitt trigger
IC2	- 74HC93 4-bit binary counter
IC3	- 74HC4051 8-channel analogue multiplexer/demultiplexer
IC4	- LM558 low-power dual-operational amplifier
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R3, R4	- 100-ohm
R2	- 10-kilo-ohm
VR1-VR8, VR10	- 10-kilo-ohm potentiometer
VR9	- 100-kilo-ohm potentiometer
Capacitors:	
C1	- 10 μ F, 16V electrolytic
C2	- 1 μ F, 16V electrolytic
C3	- 100nF ceramic disk
C4	- 10nF ceramic disk
C5-C7	- 0.33 μ F ceramic disk
C8	- 100 μ F, 16V electrolytic
Miscellaneous:	
S1	- Rotary switch
S2	- On/off toggle switch
CON1-CON6	- 2-pin terminal connector

an 8-channel analogue multiplexer/demultiplexer with three digital selected pins 9, 10 and 11. Input pins of IC3 are connected to eight potentiometers of 10-kilo-ohm each.

Variable step sizes are obtained

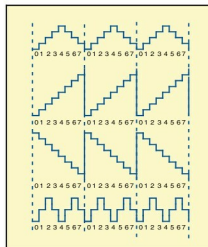


Fig. 2: Staircase output waveforms

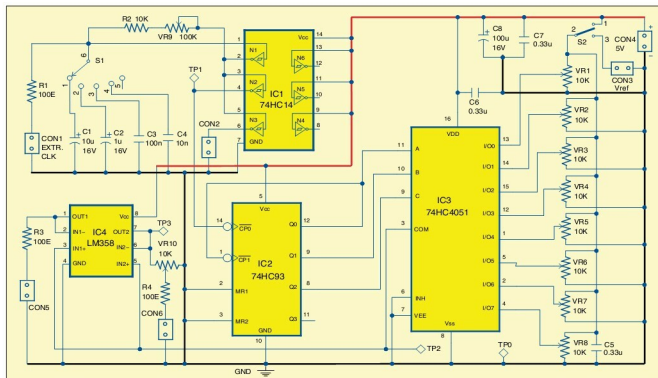


Fig. 1: Circuit diagram of simple adjustable staircase waveform generator

